

The Kernel Trick in Code

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Compare with:** Polynomial features ([Note ML-079](#)).

1. Overview

Key point: On data shaped as two circles, a linear SVM scores 55%. Switching the **kernel** (G-1004) to RBF, or to a degree-2 polynomial, scores 100%, using the same two input features.

[The kernel trick](#) showed in pictures how a kernel lifts data into a higher dimension. This Note runs it in scikit-learn on a real non-linear dataset, then looks at why no new features are needed. Figure 1 shows the four models we train. The ringed points are the **support vectors** (G-1923), the training points that fix the boundary; section 6 explains them.

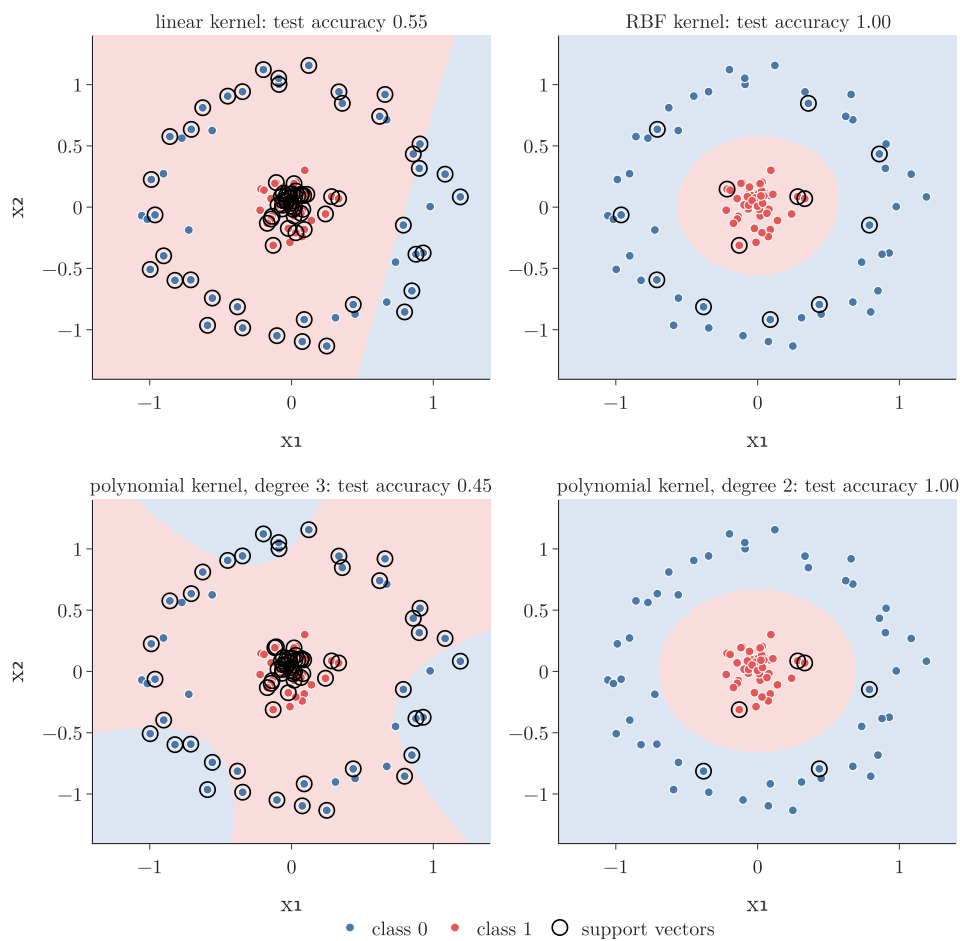


Figure 1: Decision regions of four SVMs on the circles data; ringed points are support vectors

This Note covers:

- the data: two circles (section 2);
- a linear SVM failing on it (section 3);
- lifting the data into 3D by hand (section 4);
- the RBF kernel (section 5) and the polynomial kernel (section 6);
- why the kernel is a trick: dot products without new features (section 7);
- gamma, the RBF kernel's main setting (section 8);
- an interactive playground (section 9).

All the code is in the Notebook of this Note (`ML-090-kernel-trick-code.ipynb`).

2. The data: two circles

Key point: `make_circles` generates 100 points: one class in a small inner circle, the other in a ring around it.

scikit-learn's sample generators create toy datasets. `make_circles` makes two concentric circles of points: 100 **observations** (one record each, one row of the data table), 50 per class, with two **features** (input variables, one column each): the x and y coordinates. The **target** (the output we predict) is the class. Class 1 (red) sits in the centre and class 0 (blue) in a ring around it.

Python: Generating the data.

```
from sklearn.datasets import make_circles

# factor: inner radius as a share of the outer one
X, y = make_circles(n_samples=100, factor=0.1,
                    noise=0.1, random_state=5)
X.shape          # (100, 2): x and y coordinates
```

Wherever we draw a straight line through this data, it cuts the ring in two, so no line can be a good classifier.

3. A linear SVM fails

Key point: `SVC(kernel="linear")` reaches only 55% test accuracy: a straight decision boundary cannot separate a disc from a ring.

We split the data into training and test sets, train an SVM with a linear kernel, and measure accuracy on the test set.

Python: Training a linear SVM.

```
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=0)

linear = SVC(kernel="linear")
linear.fit(X_train, y_train)
accuracy_score(y_test, linear.predict(X_test))  # 0.55
```

The accuracy is 0.55, little better than guessing, which was bound to happen with a linear classifier on non-linear data. To see the **decision boundary** (G-555), the line or curve where the model's prediction switches from one class to the other, a helper function predicts a fine grid of points and colours each by its prediction: these are the model's decision regions (see [how the model predicts](#)). Figure 1, top left, shows the result: one straight cut that leaves

many blue ring points on the red side.

4. Lifting the data into 3D by hand

Key point: Put each coordinate through $\exp(-x^2)$ and add the results as a height z . The centre points rise to about 2, the ring stays below about 1.6, and a flat plane separates them.

4.1 The transformation

Key point: $z = e^{-x_1^2} + e^{-x_2^2}$: large near the centre, small far from it.

A 2D example: [concentric circles](#) lifts these circles with $z = e^{-(x_1^2+x_2^2)}$, so the centre rises and a flat plane splits the classes. In code we use a per-coordinate variant: the bump e^{-x^2} applied to each coordinate, then added.

$$z = e^{-x_1^2} + e^{-x_2^2}$$

The per-coordinate bump behaves the same way: points near the centre, where both coordinates are small, get the largest z .

- A centre point, (0.1, 0.05):

$$z = e^{-0.01} + e^{-0.0025}$$

$$z = 0.990 + 0.998 = 1.99$$

- A ring point, (1, 0):

$$z = e^{-1} + e^0$$

$$z = 0.368 + 1 = 1.37$$

On the whole dataset, the centre points get z between 1.89 and 2.00, and the ring points between 1.01 and 1.56. So a flat plane at $z = 1.75$ separates them, and a linear classifier now works. Figure 2 animates the lift: watch the red centre points rise above the plane while every blue ring point stays below it.

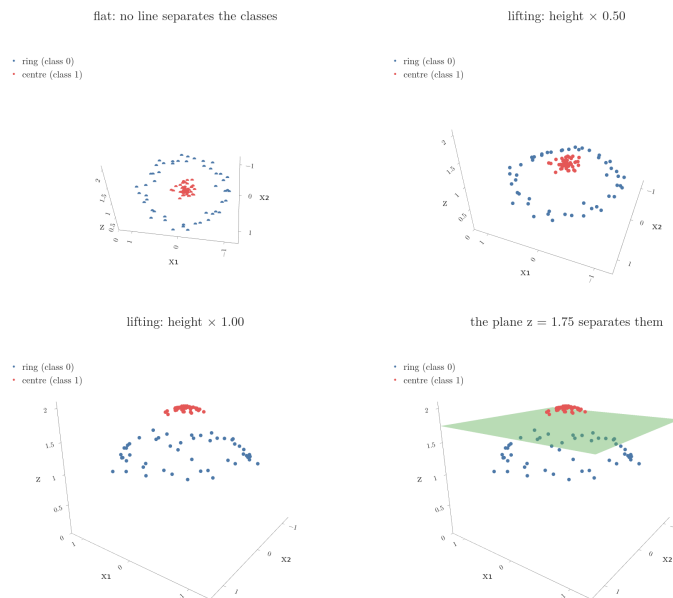


Figure 2: The circles data lifted by $z = e^{-x_1^2} + e^{-x_2^2}$: the centre class rises to 1.89–2.00, the ring stays at 1.01–1.56, and the plane $z = 1.75$ separates them

Python: The transformation and the 3D plot.

```
import numpy as np
import plotly.graph_objects as go

X_new = np.exp(-(X ** 2))      # exp(-x^2) for each coordinate
z = X_new.sum(axis=1)          # add the two columns: the height
go.Figure(go.Scatter3d(x=X[:, 0], y=X[:, 1], z=z,
                        mode="markers")).show()
```

4.2 Where to centre the bump

Key point: The bump only works if it is centred on the right class. Without knowing the centre, we would need one bump per point, which is expensive.

The lift worked because the bump e^{-x^2} is centred at 0, exactly where the inner class sits. If the inner class were centred at 5, we would need $e^{-(x-5)^2}$ instead. In higher dimensions we cannot look at the data and see where the centre is.

So SVM in effect places a bump on the data points themselves and adds the bumps up, each with its own weight; training finds the weights that separate the classes best. Doing this transformation explicitly would be very expensive as the data grows. The point of the kernel trick is that SVM never has to build these new features at all (Section 7).

Extra: The function used here, $e^{-x_1^2} + e^{-x_2^2}$, treats each coordinate separately; it is a convenient choice for the picture. The RBF kernel that SVM uses compares two points

through their distance: $K(a, b) = e^{-\gamma \|a-b\|^2}$. The SVM's decision function is a weighted sum of such bumps, one centred on each support vector: $\sum_{i \in SV} y_i \alpha_i K(x_i, x) + b$ (sklearn UG §1.4.7). Here $\sum_{i \in SV}$ means “add over every support vector i ”, and the weights α_i are found in training.

5. The RBF kernel

Key point: SVC(kernel=“rbf”) reaches 100% test accuracy, with the same two features and no feature engineering.

Now we let SVM do it. We train a new SVC with the **RBF kernel** (G-1639), `kernel="rbf"`, instead of `"linear"` and change nothing else.

Python: The RBF kernel.

```
rbf = SVC(kernel="rbf")           # "rbf" is also the default
rbf.fit(X_train, y_train)
accuracy_score(y_test, rbf.predict(X_test))    # 1.0
```

The test accuracy is 1.0. Figure 1, top right, shows its decision boundary: a closed curve around the centre class.

We did not create any extra feature: only the original two features went into the model, and the kernel did the calculation internally. Compare this with [polynomial regression](#), where we had to add a new feature for every power of the inputs.

6. The polynomial kernel

Key point: kernel=“poly” uses degree 3 by default and scores 45% here. With degree=2 it scores 100%. The degree is a hyperparameter.

The **polynomial kernel** (G-1514) is chosen with `kernel="poly"`. The polynomial kernel has an extra setting, **degree**, the degree of the polynomial. The default degree is 3, and it is ignored by all other kernels.

Kernel	Test accuracy	Support vectors (of 80)
linear	0.55	76
RBF	1.00	13
polynomial, degree 3	0.45	74
polynomial, degree 2	1.00	6

With the default degree 3, the accuracy is even worse than linear: 0.45, with the odd-shaped decision boundary of Figure 1, bottom left. With **degree=2** it jumps to 1.0 (bottom right). The degree must therefore be tuned like any hyperparameter, for example with a **grid search** (trying every combination of listed values) and **cross-validation** (scoring each setting on several held-out parts of the data; see [cross-validation with a pipeline](#)).

Python: Changing the degree.

```
for degree in [3, 2]:
    poly = SVC(kernel="poly", degree=degree)
    poly.fit(X_train, y_train)
    print(degree, accuracy_score(y_test,
                                  poly.predict(X_test)))

# 3 0.45
# 2 1.0
```

Extra: Why does degree 2 work and degree 3 fail? scikit-learn's polynomial kernel is $(\gamma a \cdot b + r)^d$ with r (`coef0`) equal to 0 by default (sklearn UG §1.4.6). With $r = 0$, the degree-3 kernel contains only terms of degree exactly 3, such as x_1^3 and $x_1^2 x_2$. A circle needs $x_1^2 + x_2^2$, a degree-2 term, which the degree-3 kernel cannot produce. The degree-2 kernel contains exactly x_1^2 , $x_1 x_2$ and x_2^2 . Setting `coef0=1` with degree 3 adds the lower-degree terms back, and in the Notebook this model also scores 1.00 on the circles.

Figure 3 shows why the degree-2 terms are enough. Watch the circle on the left turn into a straight line on the right once each coordinate is squared: in the features x_1^2 and x_2^2 the decision boundary $x_1^2 + x_2^2 = 0.34$ is linear.

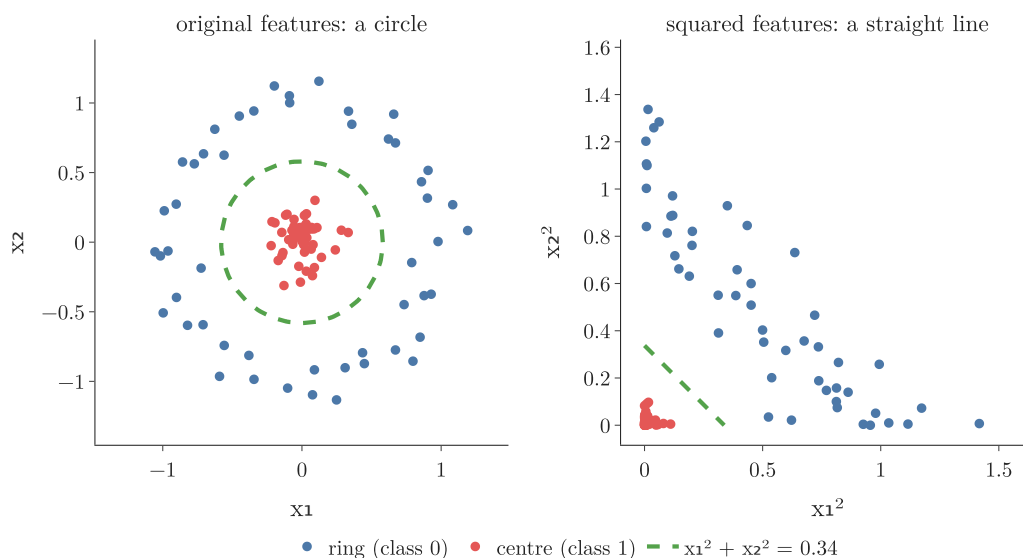


Figure 3: Left: the circles data and the circle $x_1^2 + x_2^2 = 0.34$ between the classes. Right: the same points in the squared features, where that circle is the straight line $x_1^2 + x_2^2 = 0.34$

The table also shows the **support vectors** (G-1923). The linear model needs almost every training point; the good kernels need only a few. The reason: every point that sits on the margin, inside it or on the wrong side of the boundary becomes a support vector (ESL §12.2.1). A straight line fits the circles badly, so most points end up inside its margin; a curved boundary that fits well leaves only a few points near it.

Extra: The Notebook checks the margin rule for the linear model. The linear model misclassifies 30 of the 80 training points, and exactly 76 points have $y \cdot f(x) < 1$ (inside the margin or on the wrong side): the same 76 as its support vectors.

7. Why it is a trick

Key point: A kernel gives the dot product of two points in the higher-dimensional space directly from the original coordinates, so the new features are never built.

SVM's training and predictions only need **dot products** (G-634; multiply matching entries of two vectors and add, see [computing the dot product](#)) between pairs of points (MML §12.4). A kernel is a function $K(a, b)$ that returns the dot product the two points would have **after** the transformation, without carrying the transformation out. We put the original coordinates into the kernel formula, and out comes the value we need.

As a formula, for the degree-2 polynomial kernel the explicit features are:

$$\phi(x) = (x_1^2, \sqrt{2}x_1x_2, x_2^2)$$

The kernel is:

$$K(a, b) = (a \cdot b)^2 = \phi(a) \cdot \phi(b)$$

With numbers, for $a = (1, 2)$ and $b = (3, 1)$:

- Kernel:

$$a \cdot b = 3 + 2 = 5$$

$$K(a, b) = 5^2 = 25$$

- Explicit features:

$$\phi(a) = (1, 2\sqrt{2}, 4)$$

$$\phi(b) = (9, 3\sqrt{2}, 1)$$

$$\phi(a) \cdot \phi(b) = 9 + 12 + 4 = 25$$

Figure 4 draws the two routes side by side; watch both arrive at 25, while only the top route builds new features.

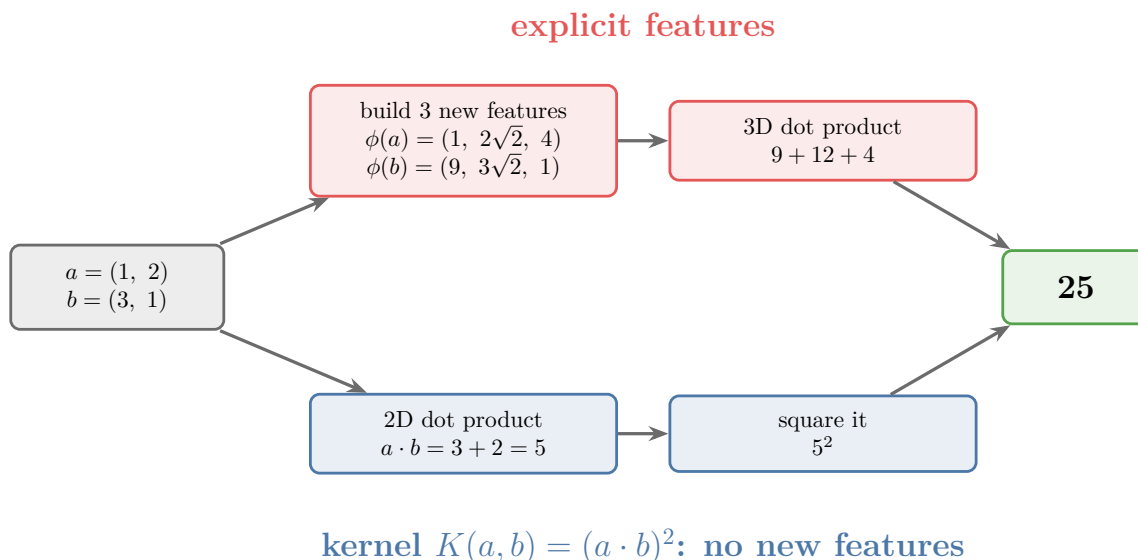


Figure 4: Two routes to $\phi(a) \cdot \phi(b)$ for $a = (1, 2)$ and $b = (3, 1)$. Top: build the three degree-2 features, then take a 3D dot product. Bottom: the kernel, a 2D dot product squared

The kernel needed one 2D dot product and a square; the explicit route needed three new features per point. With more inputs and higher degrees the explicit features multiply quickly, while the kernel stays one dot product. For the RBF kernel the matching feature space is infinite-dimensional, so building it would be impossible, yet the kernel value is cheap (MML §12.4). Getting the dot product without building the features is exactly the **kernel trick** (G-1008).

8. Gamma: how far one point's influence reaches

Key point: The RBF kernel value between two points falls as their distance grows. Gamma sets how fast. Small gamma: far points still count, smooth boundary. Large gamma: only very close points count, and the boundary wraps around single points.

The RBF kernel compares two points through their distance:

$$K(a, b) = e^{-\gamma \|a - b\|^2}$$

Here $\|a - b\|$ is the distance between points a and b . For $a = (1, 2)$ and $b = (4, 6)$:

$$a - b = (-3, -4)$$

$$\|a - b\| = \sqrt{(-3)^2 + (-4)^2}$$

$$\|a - b\| = \sqrt{9 + 16} = 5$$

With $\gamma = 0.1$, the kernel value of these two points is:

$$K(a, b) = e^{-0.1 \times 5^2} = e^{-2.5}$$

$$K(a, b) = 0.08$$

The value is 1 when the two points coincide and falls towards 0 as they move apart. We can read the value as the **influence** one training point has on another point: close points have a lot of influence, far points almost none. The setting **gamma** (G-823), γ , multiplies the squared distance, so gamma decides how quickly the influence dies out.

For two points at distance 1, $K = e^{-\gamma}$:

gamma (G-823)	K at distance 1	Reading
0.1	$e^{-0.1} = 0.90$	a point at distance 1 still has almost full influence
1	$e^{-1} = 0.37$	about a third
10	$e^{-10} = 0.00005$	practically none

Figure 5 draws the kernel value against the distance for the three gammas. Watch the red curve (gamma = 10): the influence is gone before distance 1, while the green curve (gamma = 0.1) is still about 0.4 at distance 3.

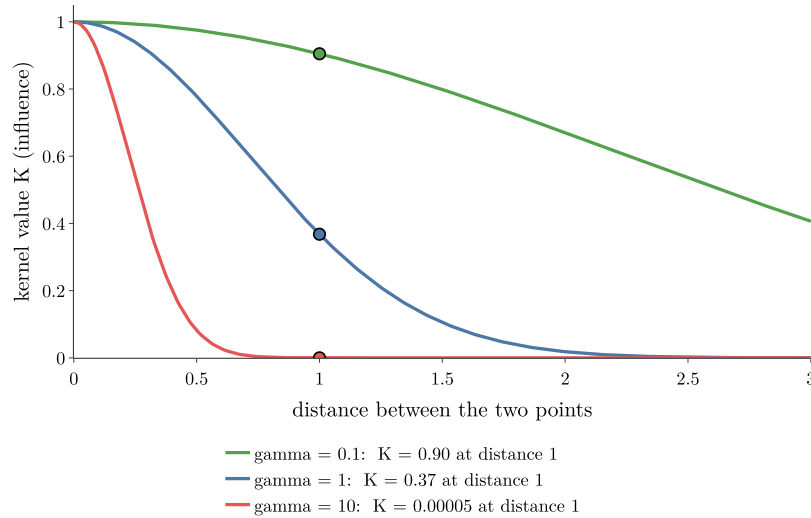


Figure 5: The RBF kernel value against the distance between two points, for $\gamma = 0.1$, 1 and 10. The dots mark distance 1: 0.90, 0.37 and 0.00005

A small gamma makes each bump wide, so many training points have a say at every location and the decision boundary is smooth (risk of underfitting). A large gamma makes each bump narrow, so only the nearest training points have a say and the boundary wraps around individual points (risk of overfitting). Gamma is a **hyperparameter** (G-910).

Figure 6 runs a gamma sweep on the moons data (two interleaved half-circles of points), from 0.01 to 1000, with $C = 1$ (the setting that trades a wide margin against training mistakes, see [the hyperparameter C](#)). Watch the decision boundary go from an almost straight cut to a curve that follows the moons, then break into small islands around single points. Training accuracy climbs to 1.00, while accuracy on 5,000 fresh points from the same generator peaks at 0.97 (gamma about 5) and falls to 0.68 at gamma 1000.

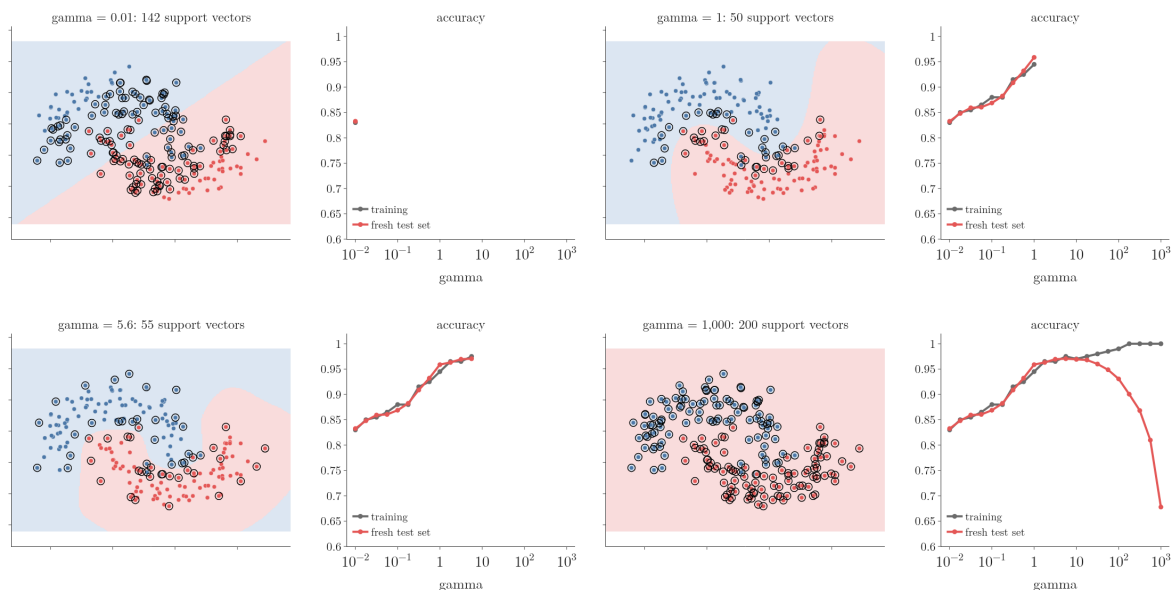


Figure 6: RBF SVM on the moons data as gamma grows. Left: decision regions, with support vectors ringed. Right: training accuracy and accuracy on a fresh test set of 5,000 points

Extra: scikit-learn's default, `gamma="scale"`, sets gamma from the spread of the data: $1/(p \times \text{variance of } X)$, where p is `n_features`, the number of features. `C` and gamma are usually tuned together with a grid search (sklearn, "RBF SVM parameters").

9. An interactive playground

Key point: `app.py` lets you change the dataset, kernel, `C`, gamma and degree and watch the decision regions and support vectors update.

The folder of this Note contains `app.py`, a small Dash app. Run `python app.py` and open <http://127.0.0.1:8050>. Figure 7 shows it with its default settings.

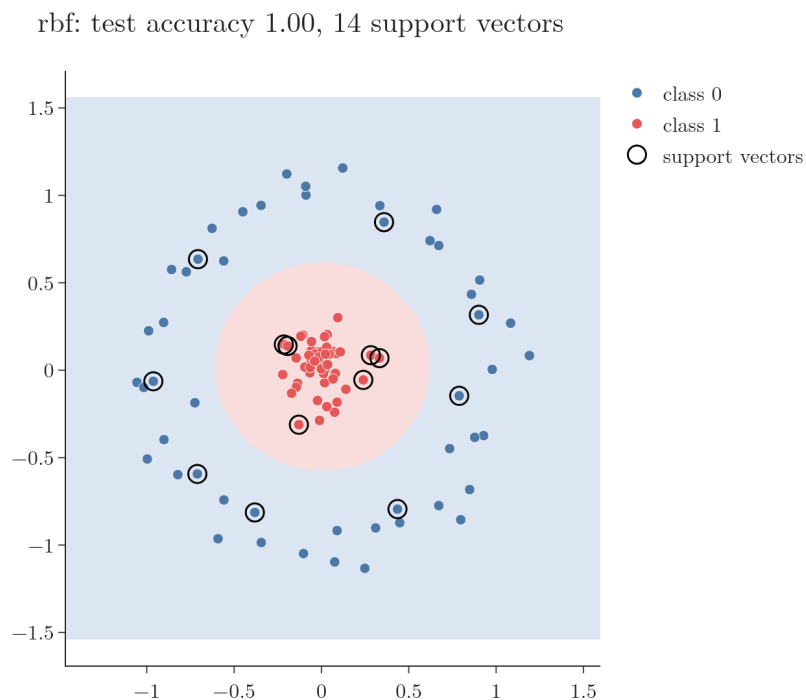


Figure 7: The playground on the circles data with the RBF kernel, $C = 1$ and $\gamma = 1$

Things to try:

- **linear** on the circles: one straight cut, about 0.55 accuracy, and nearly every point a support vector.
- **poly** with degree 3, then 2: the accuracy jumps from 0.45 to 1.00.
- **C** from 3 down to -3 (that is, 10^3 to 10^{-3}): small C widens the margin and adds support vectors, as in [the hyperparameter C](#).
- **gamma** with the RBF kernel on the moons data: from smooth boundaries at small γ to tight islands around single points at large γ , as in Figure 6.

10. Summary

Model	Code	Test accuracy
Linear SVM	<code>SVC(kernel="linear")</code>	0.55
RBF kernel	<code>SVC(kernel="rbf")</code>	1.00
Polynomial, degree 3 (default)	<code>SVC(kernel="poly")</code>	0.45
Polynomial, degree 2	<code>SVC(kernel="poly", degree=2)</code>	1.00

- A linear SVM cannot separate circular classes.
- Lifting with $z = e^{-x_1^2} + e^{-x_2^2}$ makes them separable by a plane in 3D.
- With a kernel, SVM gets the same effect from the original two features: no new features built.
- The kernel's settings (degree, gamma) and C are hyperparameters to tune.
- Gamma sets how far one point's influence reaches: small gamma gives smooth boundaries, large gamma tight ones.
- A kernel returns the dot product in the higher-dimensional space directly: the kernel trick.

11. Sources

Built from

- CampusX, “Kernel Trick in SVM | Code Example”, YouTube, <https://www.youtube.com/watch?v=pjv>
- StatQuest with Josh Starmer, “Support Vector Machines Part 3: The Radial (RBF) Kernel (Part 3 of 3)”, YouTube, https://www.youtube.com/watch?v=Qc5IyLW_hns (gamma scales the squared distance, and so the influence two points have on each other; section 8)

Other references

- **ESL:** T. Hastie, R. Tibshirani and J. Friedman, *The Elements of Statistical Learning*, 2nd ed., Springer, 2009. Section 12.2.1.
- **MML:** M. P. Deisenroth, A. A. Faisal and C. S. Ong, *Mathematics for Machine Learning*, Cambridge University Press, 2020. Section 12.4, Kernels.
- **sklearn UG:** scikit-learn User Guide, Section 1.4, Support Vector Machines (1.4.6 Kernel functions, 1.4.7 Mathematical formulation), and the **SVC** reference page. scikit-learn.org/stable/modules/svm.html
- **sklearn, “RBF SVM parameters”:** scikit-learn example gallery, *RBF SVM parameters*. [scikit-learn.org, auto_examples/svm/plot_rbf_parameters](https://scikit-learn.org/auto_examples/svm/plot_rbf_parameters)

12. Key terms

Term	Meaning
make_circles (G-110)	A scikit-learn generator of two concentric circles of points, a standard non-linear test dataset
Feature map (ϕ) (G-765)	The explicit transformation of a point into the higher-dimensional space
Kernel function $K(a, b)$ (G-1006)	A function that returns $\phi(a) \cdot \phi(b)$ directly from the original points
degree	The degree of SVC's polynomial kernel; default 3
gamma (G-823)	How far one point's influence reaches in the RBF kernel; large gamma gives tighter boundaries